Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/09/2004
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Project Title:	Behavioral And Neurochemical Effects Of Exposure To Heavy Particles		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
Start Date:	06/15/2003	**End Date:**	06/14/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG9-1529		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Exposing rats to non-lethal doses of 56Fe particles produces changes in the central nervous system, particularly in dopaminergic function. Dopamine is a neurotransmitter that mediates a wide range of behaviors, including motor behavior, some types of learning, reinforcement, sensory information processing, food intake and the production of abnormal behavior patterns. Because changes in each of these behaviors can affect the performance of astronauts, the knowledge of the behaviors affected by exposure and an understanding of the neural mechanisms mediating the behaviors is essential. The proposed research program is designed to gain additional information on alterations in brain neurochemistry and behavior following exposure to particles of high energy and charge, such as those that might be encountered during space travel outside the magnetic field of the earth. The research is designed to further describe and characterize behavioral and neurochemical deficits produced by exposure to heavy particles, determine their underlying mechanisms, and develop approaches to minimize such deficits. In addition, the research program proposes to examine factors such as age and diet that may interact with the effects of exposure to heavy particles, and which may, in turn,		

	affect the health and performance of astronauts during exploratory missions to other planets.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research which we have conducted previously has shown that exposing young rats (less than 3 mo. old) to low doses of heavy particles (56Fe, 600 MeV/n) has shown that the neurochemical deficits produced by this exposure are similar to those that are observed in aged rats (24 mo. old). The research which we have completed at BNL indicates that similar deficits are observed following exposure to one GeV/n 56Fe particles as well. Thus, exposing rats to low doses of 56Fe particles provides a way to accelerate aging in experimental animals with respect to certain brain and behavioral parameters so that experimenters do no have to wait for 24 months to obtain old animals. The research program is designed to understand the mechanisms by which exposure to heavy particles (primarily 56Fe) produce their effects on brain and behavior. Although the impetus for the research is to understand and minimize the effects of exposure to heavy particles on astronauts, the research program necessarily has implications for the understanding of the natural aging process. Because exposure to 56Fe particles may produce accelerated aging, this research is also indirectly concerned with the basic processes underlying the biology of aging. Similarly, because the ultimate goal of the research program is to develop interventions to minimize the effects of exposure to heavy particles on astronauts, the interventions may also prove useful with the natural aging process.
Task Progress:	The experiments covered in the current report were conducted during a single run at BNL in Nov. 2003. Two experimental groups were run: (1) young S-D rats were exposed to 48Ti particles to provide a comparison with the effects of exposure to 56Fe particles; and (2) 7-month old and 17-month old F-344 rats were exposed to 56Fe particles to study the effects of age on the cognitive/behavioral effects of exposure. Two specific hypotheses about the effect of exposure to HZE particles on behavior were tested: 1. Exposing rats to 48Ti particles (1.2 GeV/n), which has a lower LET than 1 GeV/n 56Fe particles, will require higher doses to affect behavioral performance. 2. Exposing 7-month and 17-month old rats to 56Fe particles with disrupt performance at lower doses than required to affect the performance of young rats. Effects of Exposure to 48Ti Particles on Behavior CTA Learning A conditioned taste aversion (CTA) is produced when a novel taste (10% sucrose solution) is paired with a novel unconditioned stimulus (amphetamine, 3 mg/kg) such that the organism will avoid ingestion of that solution at a subsequent presentation. Exposure to lower doses of 56Fe particles (600, 1000 or 5000 MeV/n) disrupts the acquisition of the dopamine-mediated CTA produced by injection of amphetamine whereas exposure to higher doses enhances the avoidance of the normally preferred sucrose solution. The threshold for the disruption of the CTA parallels the particle LET, being lowest for 600 MeV/n 56Fe particles (which has the highest LET) and increases as particle energy increases and particle LET decreases. . Based on previous research, the current run using 48Ti particles (1.2 GeV/n, LET ˜ 114 keV/µm) examined the effects of exposure to doses (0.8 - 2.0 Gy) on the acquisition of the dopamine-mediated CTA produced by injection of amphetamine. Exposure to 0.8 - 1.5 Gy did not affect the acquisition of a CTA. Because the 2.0 Gy dose did seem do produce a greater aversion to the sucrose solution than seen in the control animals, the inference is that the doses of 48Ti were too high. Subsequent studies must evaluate the effects of lower doses of 48Ti particles on the acquisition of an amphetamine-induced CTA. Operant Responding Operant (instrumental) conditioning occurs where the organism learns to make a response to obtain reward or avoid punishment. It is broadly construed to include all forms of complex learning. The specific task involved training the animals to respond on an ascending fixed-ratio reinforcement schedule. In a fixed-ratio design, the animal is rewarded after n number of responses; such that on an FR-5 schedule, the rat is rewarded after 5 responses (bar presses) and on an FR-10 schedule, after 10 responses. Previous research has shown that the threshold for the disruption of operant responding by exposure to 56Fe particles (1 GeV/n) is 1.5-2.0 Gy. The results of the present experiment with 48Ti particles shows no disruption of responding in young rats exposed to 1.0 or 1.5 Gy. This suggests that the threshold for an effect of exposure to 48Ti on this particular endpoint is somewhat higher than that following exposure to 56Fe particles. As such, these results may be consistent with the hypothesis that the relative effectiveness of heavy particles in disrupting performance is correlated with particle LET. However, there was an apparent increase in response rate in the rats exposed to 2.0 Gy of 48Ti particles. While the increase in responding was not statistically significant, it does suggest that effects of exposure may not be linear; i.e., may not produce a uniform decrement in performance. However, in the absence of additional data points, it is not possible to validity of these results. Therefore, future runs at the NSRL utilizing 48Ti particles must evaluate the effects of a wider range of doses on a range of behavioral endpoints. Effects of Age and Radiation on Behavior Previous research has found that many of the behavioral deficits that characterize the aged organism can be produced by exposure to heavy particles. In addition, this research has shown that as irradiated rats age, there is a decrement in performance that suggests an interaction between exposure to 56Fe particles and the age of testing. Because astronauts are likely be middle-aged, it is essential to determine whether the age of the individual can influence the effects of exposure to heavy particles on organisms of different ages. The present experiments represent an initial evaluation of the direct effects of exposure to 56Fe particles on performance in organisms of different ages. CTA Learning Exposing 7-month old F-344 rats to 0.5 Gy of 56Fe particles, produces a disruption of amphetamine-induced CTA learning. This dose is lower than needed in younger S-D rats, for which a dose of 0.8 Gy is needed to produce a disruption of the dopamine-mediated CTA. Increasing the dose to 1.5 Gy, has no further effects on CTA learning in the 7-month old rats. The data are less clear with the older (17-month) rats, in which a dose of 1.5 Gy is needed to produce a significant increase in sucrose preference. Operant Responding When 7-month old rats (9 months old at the time of testing) are exposed to 0.5 of 56Fe particles, there is a reduction in response rate at ratios =FR-15. These results are in contrast to the results obtained with young S-D rats in which exposure to 1.5 - 2.0 Gy is needed in order to affect responding. The effects of exposure of on the performance of the 7-month old rats did not vary as a function of dose - increasing the dose of 56Fe particles to 1.5 Gy did not produce a further deterioration in performance.

	The results obtained with the 19-month old rats (17 months old at the time of irradiation) were not as predicted. While the rats exposed to 0.5 Gy showed a somewhat decreased level of performance compared to the non-irradiated control rats, the rats exposed to 1.5 Gy of 56Fe particles showed an enhanced level of performance. The mechanisms that may be responsible for this effect are not immediately obvious. First of all, these results must be replicated in order to ascertain their validity. However, it is interesting to note that exposing rats to 2.0 Gy of 48Ti particles (the highest dose tested) also produced a slight enhancement of performance. Water Maze Performance The Morris water maze is the standard procedure used to evaluate age-related changes in spatial performance. The task requires an animal to use spatial learning and memory to find a hidden platform just below the surface of the water, and to remember its location from the previous trial. The rat must acquire a cognitive representation of location in space and have the ability to effectively navigate the environment; therefore, the animal must use distal cues to effectively locate the hidden platform. Accurate navigation is rewarded by escape from the pool. Preliminary analyses confirmed the effects of age on performance in the Morris Water Maze, but failed to show a clear interaction between exposure to radiation and age at the time of exposure. There were significant effects of exposure on the performance of the 9-month old rats on performance during the probe trials without the platform present. However, similar effects were not observed in the older animals (19 months old at the time of testing). These results are similar to those obtained with the CTA and instrumental performance. For both of those behaviors, as well as water maze performance, the major effects of irradiation were observed with the middle-aged rats. It is possible that the performance of the older animals is already so poor that the additional effects produced by exposure to 56Fe particles can not be expressed in behavior. However, evaluation of this hypothesis will require additional experiments using animals of different ages exposed to a range of doses.

Bibliography Type:	Description: (Last Updated: 05/08/2025)
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